

Recurrent models typically factor computation along with symbols or tokens across time to process sequences. This inherent sequential nature precludes parallelization within training examples, which becomes critical for longer sequences. In fact, as memory constraints limit batching across examples.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models for various tasks, allowing modeling of dependencies without regard to their distance from input or output sequences.

In this work we propose the Transformer, model architecture eschewing recurrence instead relying entirely on an attention mechanism to relate dependencies between input and output.